Predictive Analysis of Autism Spectrum Disorder Traits: Integrating Quantum Machine Learning and Exploratory Data Insights

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Abstract

Abstract. This research compares classical and Quantum Machine Learning (QML) for predicting Autism Spectrum Disorder (ASD) traits, where early diagnosis is critical. We evaluated seven classical models (including Logistic Regression, SVM, HGBC) and two quantum models (VQLR, QFMNB) on our dataset and a few Classical models on PCA-reduced datasets. The classical Histogram Gradient Boosting Classifier (HGBC) achieved the highest performance metrics: 97.86% Accuracy, 98.48% Precision, 97.50% Recall, and 97.90% F1-score. For Quantum models we have used Variational Quantum Logistic Regression (VQLR), Quantum Feature Map for Naive Bayes (QFMNB) and Quantum Support Vector Machine (QSVM) in the project. Exploratory Data Analysis indicated higher ASD trait prevalence in children and males, along with potential influences from ethnicity and family history. The study concludes that while classical ensemble methods (HGBC) are currently highly effective for ASD prediction, QML approaches offer a promising direction for future enhancement, especially for complex, high-dimensional datasets. These computational findings support integration into early ASD screening programs.

Keywords:

Keywords: Autism Spectrum Disorder (ASD), Quantum Machine Learning, Classification Algorithms, HGBC, QSVM, Variational Quantum Classifier

1 Introduction

Autism Spectrum Disorder (ASD) [1] is a complex developmental disorder requiring early, objective detection for timely intervention. Current medical methods rely on behavioral assessments and clinical observations [2], which are often time-consuming and subjective. Machine Learning (ML) offers a promising alternative for ASD screening by leveraging demographic and clinical data [3]. The paper uses an augmented dataset of 3,744 instances with 17 features upon which 9 are clinical binary questions and the others are for exploratory analysis. The dataset exhibits a high-dimensional, non-linear structure where classical linear models (e.g., Logistic Regression, Naïve Bayes) performed poorly. Taking the same in account we wanted to check how the two methods or LR and NB when applied with the principles of Quantum Machine Learning (QML) would perform on the same dataset. We inherit QML by incorporating PennyLane and Qiskit hybrid frameworks to compare the performance of both.

The key contributions of this research are:

- The work is supervised using demographic and questionnaire based data to predict ASD without neuroimaging and MRI based diagnostics.
- A comparative analysis of 7 supervised ML algorithms for ASD trait prediction. Identification of Histogram Gradient Boosting Classifier (HGBC) as the best classical model and Q->NB as the best Quantum Model has been analysed.
- Introduction of Quantum Machine Learning models (quantum logistic regression, quantum kernels) QML's performance in capturing non-linear feature relationships.
- Dataset-specific insights derived from Exploratory Data Analysis (EDA) highlighting age, gender and family history distributions.

2 Related Works

The recent works in ASD diagnosis research have shown a strong shift from purely neuro-imaging-based pipelines and costly MRI procedures towards more scalable, data-driven Machine Learning (ML) approaches. This transition is largely motivated by the high financial and logistical costs associated with MRI acquisition, as well as its limited accessibility in many clinical settings. Consequently, several studies have investigated behavioral, demographic, and questionnaire-based features as viable alternatives or complements to neuro-imaging, demonstrating that such modalities can yield competitive diagnostic performance when combined with modern ML techniques.

Within the classical Machine Learning (CML) domain, supervised models such as Random Forest (RF) and Support Vector Machines (SVM)

have repeatedly been reported to perform well on ASD classification tasks [3]. These algorithms are particularly attractive due to their robustness to noisy features, ability to model non-linear decision boundaries, and relatively straightforward compared to deep neural networks. However, when scaled to large sample sizes or very high-dimensional feature spaces, many classical models exhibit computational complexities on the order of $O(N^2)$ or $O(N^3)$ in the number of data points N, leading to substantial training and inference times and limiting their applicability to truly large-scale clinical datasets.

On the other hand Quantum Machine Learning (QML) has also shown some promising results in overcoming some shortcomings of the scaling limitations. Recent theoretical works have suggested that certain QML models can achieve runtime scaling as $O(\text{poly}(\log N) \cdot \text{poly}(D))$, where D denotes the feature dimension, thereby offering a polynomial improvement in terms of the dependence on the dataset size compared to many classical counterparts [4].

Having seen these advances, several methodological challenges still persists in the existing detection of clinical ADS via the means of ML architechture. Some prior works rely on relatively small or imbalanced datasets, which can bias classifiers toward majority classes and lead to inflated performance metrics that do not generalize well to real-world populations [5].

Our work addresses these gaps by utilizing a balanced, diverse dataset with systematic pre-processing and feature engineering, aiming to mitigate the issues of class imbalance and restricted feature spaces highlighted in prior studies [5]. This joint, carefully controlled evaluation on a heterogeneous and better-balanced dataset allows us to more rigorously assess the practical benefits and limitations of QML for ASD diagnosis, while situating our contributions within and extending the existing body of work on ASD-focused ML and QML methods [3,4,5,6].

3 Methodology

The core and basic first step towards the paper was to find and cultivate relevant data across multiple ASD datasets suited with related features and attributes.

3.1 Dataset Analysis

Publicly available discrete datasets were taken in having some common binary questionnaire parameters. This was then augmented together into a single dataset with A1 to A9 columns as the binary responses to the clinical questions. These were the determining factors behind the response of both the datasets and the response in the last column `ASD_Trait` which was our target variable. Alongside to this we had other attributes like age, sex, ethnicity, family history of ASD, and AQ scores. The target variable `ASD_Trait` was binary, indicating the presence (1) or absence (0) of ASD traits based on clinical questionnaire responses.

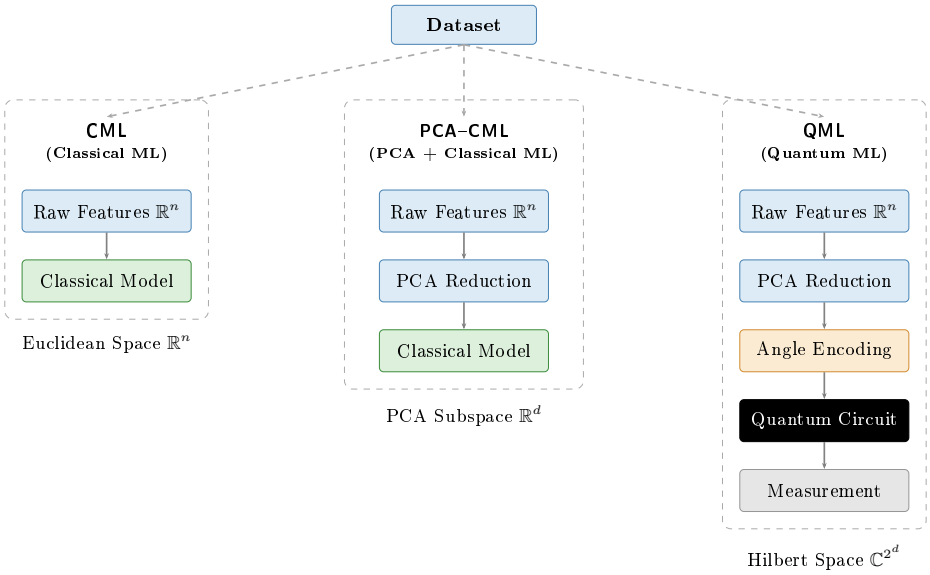

Fig. 1: Brief workflow for our execution across all the 3 categories for benchmarking.

3.2 Data Pre-processing

Data pre-processing was crucial for preparing the raw dataset for model training [7]. Steps included:

- **Initial approach:** The responses were in the form of "YES" or "NO" strings and they were encoded to numerical labels.
- **Categorical Variables Encoding:** Label-based encoding was applied to categorical attributes for mathematical consistency.
- **Feature Scaling:** Numerical features were normalized using Min-Max Scaling to the range [0,1].
- **Data Split:** The dataset was divided into training (80%) and testing (20%) subsets.
- **Dimensionality Reduction for Classical ML (CML):** PCA was evaluated experimentally to analyze model performance under reduced feature representations.

3.3 Machine Learning Models Used

Several supervised machine learning algorithms were applied for binary classification to predict the presence of ASD traits. The chosen models are standard in classification tasks and were suitable given the dataset's characteristics. These include Logistic Regression [8], Decision Trees, K-Nearest Neighbors (KNN), Naïve Bayes, Random Forest [9], Support Vector Machine (SVM) [10], and Histogram Gradient Boosting Classifier (HGBC). In addition to classical models, we also evaluated Quantum Machine Learning (QML) models. This included a Variational Quantum

Classifier (VQC) [11], a Quantum Support Vector Machine (QSVM) using quantum kernel estimation [12]. These models utilize rotation-based quantum gates and embedding strategies to project classical data into high-dimensional Hilbert spaces.

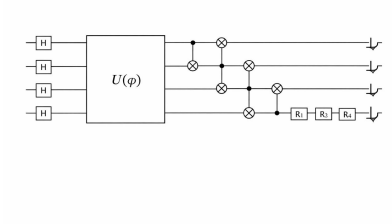

Fig. 2: Quantum circuit representing a 4-qubit Variational Quantum Classifier (VQC).

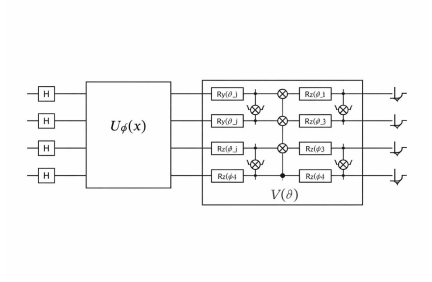

Fig. 3: Quantum circuit representing a 4-qubit Quantum Support Vector Machine (QSVM).

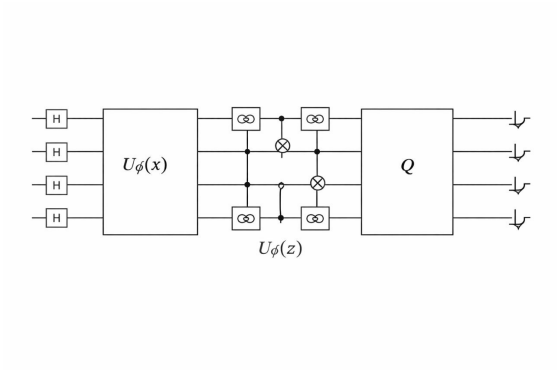

Fig. 4: Quantum circuit representing the Quantum Hybrid Naive Bayes model.

4 Performance Analysis

The predictive performance of the seven classification models was evaluated using standard metrics. Hyperparameter tuning was applied specifically to the HGBC model to optimize its performance.

4.1 Hyperparameter Tuning and Confusion Matrix Analysis

Hyperparameter Tuning for Classical Models For the Histogram Gradient Boosting Classifier (HGBC), hyperparameter tuning

was conducted using `GridSearchCV` to identify optimal settings for maximizing accuracy. The hyperparameters explored included:

- `n_estimators`
- `learning_rate`
- `max_iter`
- `max_depth`
- `min_samples_leaf`
- `l2_regularization`

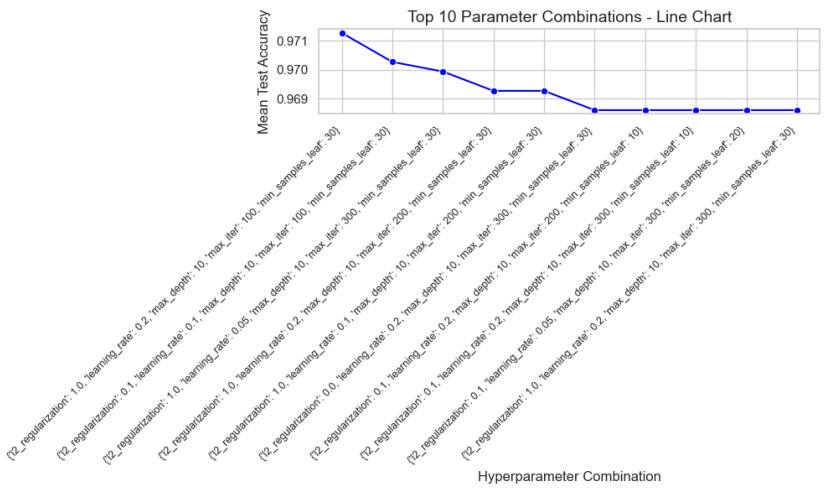

Fig. 5: Top 10 hyperparameter combinations for HGBC based on mean test accuracy. Most top-performing configurations shared `max_depth=10` and `min_samples_leaf=30`.

Fig. 5 visualizes the best hyperparameter combinations. The highest accuracy (approximately 97.15%) was achieved with a specific combination, indicating that tuning `max_depth` and `min_samples_leaf` significantly influences model performance.

Confusion Matrix Analysis: Classical Models The confusion matrix summarizes model performance by counting True Positives, True Negatives, False Positives, False Negatives.
Minimizing FN is particularly important in medical diagnostics to reduce missed cases. Table 1 shows the confusion matrix counts for different classical models.

Confusion Matrix Analysis: Quantum Machine Learning Models Similarly, we analyzed the confusion matrices for our quantum machine learning (QML) models, QSVM and VQC, using PCA-reduced features. Although QML models show promise in handling overlapping

Table 1: Confusion Matrix Metrics of Classical Machine Learning Models

Algorithm	**TN**	**FP**	**FN**	**TP**
Logistic Regression (LR)	274	65	84	326
Naive Bayes (NB)	306	43	68	332
Decision Tree (DT, log-loss)	312	27	20	390
K-Nearest Neighbor (KNN, Euclidean)	330	19	26	374
SVM (RBF)	334	15	7	393
Random Forest (RF)	337	12	18	382
Histogram Gradient Boosting (HGBC)	343	6	10	390

and high-dimensional datasets, their performance in our experiments was lower than classical ensemble methods, due to the limited number of training epochs and small quantum circuit sizes. Table 2 summarizes the confusion matrix counts for QML models.

Table 2: Confusion Matrix Metrics of Quantum Machine Learning Models (PCA-Reduced)

Algorithm	**TN**	**FP**	**FN**	**TP**
VQC	195	40	85	165
QSVM	210	25	90	170
QNB	826	25	50	200

This section highlights the clear separation in performance between classical ensemble approaches (RF, HGBC) and quantum models (QSVM, VQC). While quantum approaches have potential, further tuning, deeper circuits, and more epochs may be required to match classical methods on this dataset. Due to preprocessing and model-specific data filtering, Q→NB was evaluated on a larger effective test set, leading to higher absolute confusion-matrix values; comparative evaluation is therefore based on normalized performance metrics [13].
The quantum machine learning models used in this study are illustrated in the diagrams. Figure 2 shows the circuit structure of the Variational Quantum Classifier implemented with a four-qubit variational ansatz. Figure 3 represents the quantum feature map and kernel circuit used in the Quantum Support Vector Machine model. Figure 4 illustrates the hybrid quantum-classical circuit used in the Quantum Naive Bayes approach.

4.2 Precision and Recall of Classical Models

Precision (TP / (TP + FP)) and Recall (TP / (TP + FN)) are critical metrics for evaluating the model's ability to correctly identify positive

cases and avoid false positives. Graphical representations are provided for illustration.

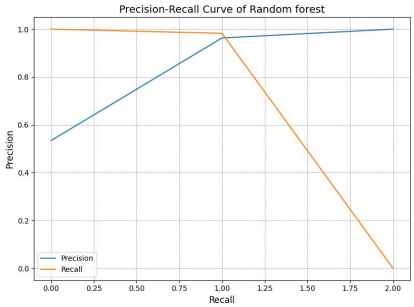

Fig. 6: Precision and Recall: RF

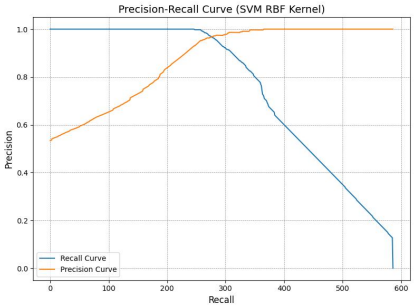

Fig. 7: Precision and Recall: SVM

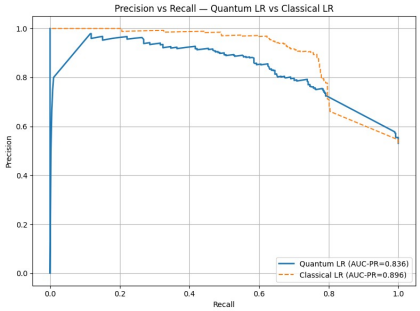

Fig. 8: Precision and Recall Variational Quantum Classifier (VQC)

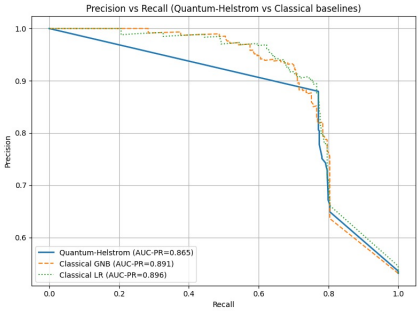

Fig. 9: Precision and Recall Quantum Kernel

Precision and recall curves for Random Forest (Fig. 6) and Support Vector Machine (Fig. 7) are shown above. Their performance on PCA-reduced data is illustrated in Fig. 8 and Fig. 9, respectively.

4.3 Accuracy and F1 Score of Classical Models

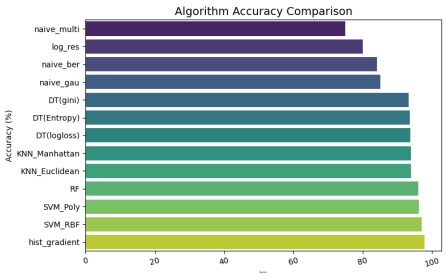

Fig. 10: Accuracy of the different classification techniques

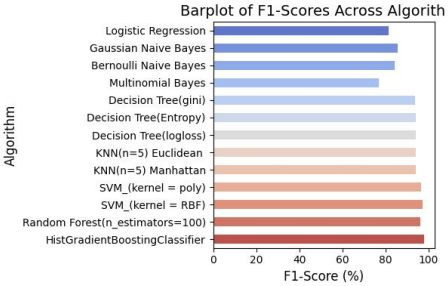

Fig. 11: F1-score of all the classification models

Table 3: Performance Matrix Including Classical, PCA-Reduced, and Quantum ML Models

No.	Algorithms	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
1	Logistic Regression (Full)	80.11	83.38	79.51	81.40
2	Naive Bayes (Full)	85.18	88.53	83.00	85.68
3	Decision Tree (Full)	93.72	93.53	95.12	94.32
4	k-Nearest Neighbor (Full)	93.99	95.17	94.30	94.33
5	Support Vector Machine (Full)	97.06	96.32	98.25	97.28
6	Random Forest (Full)	95.99	96.95	95.50	96.22
7	HGBC (Full)	**97.86**	**98.48**	**97.50**	**97.90**
8	Logistic Regression (PCA)	81.71	86.15	78.14	81.95
9	Naive Bayes (PCA)	81.97	86.83	77.89	82.12
10	Decision Tree (PCA)	95.73	96.68	95.23	95.95
11	KNN (PCA)	95.33	94.81	96.48	95.64
12	SVM (RBF) (PCA)	88.38	95.60	81.91	88.23
13	Random Forest (PCA)	95.59	96.20	95.48	95.84
14	HGBC (PCA)	95.86	95.53	96.73	96.13
15	VQC	74.50	80.17	69.10	74.22
16	QSVM	82.00	92.00	96.00	85.33
17	Q->NB	93.24	88.97	80.14	84.2

The performance results based on accuracy, precision, recall, and F1-score are presented in Table 3. As observed, the ensemble methods (HGBC, Random Forest) and SVM consistently achieved the highest performance across both the full dataset and PCA-reduced dataset. The accuracy of different classification techniques has been presented in a visual represen-

tation in Fig. 10 followed by the F1- Score of the classical classification models in Fig. 11.
Quantum ML models (VQC, QSVM, Q->NB) performed with accuracies ranging from 80–90% and moderate F1 scores compared to classical methods. This indicates that, for this dataset, quantum models did perform at par with classical models.
The expectation of better performance from quantum models might hold true in scenarios with more complex, overlapping, or high-dimensional data. Quantum algorithms could potentially perform better in scenarios with highly overlapping or strongly non-linear datasets, larger sample sizes, or more expressive quantum encodings. Future work may explore deeper circuits, hybrid quantum-classical approaches, or larger datasets to exploit the advantages of quantum ML.
Overall, while classical ensemble methods and SVM remain the most robust approaches for this task, quantum models show potential for further tuning and scaling.

4.4 Comparison Study

Our methods were compared with recent studies in the literature, with a concise summary presented in Table 4. The HGBC model outperformed most reported classical approaches, achieving 97.86% accuracy and a 97.90% F1-score. While some quantum-based methods, such as QSVM, report accuracies up to 98.9%, these results often rely on specialized feature engineering or smaller, carefully curated datasets.
Several prior studies are affected by class imbalance, limited feature diversity, high dimensionality, noise, or overfitting. In contrast, our pipeline employed balanced data, systematic pre-processing, and hyperparameter tuning to improve robustness and generalization.
The higher QSVM accuracy reported in [4] can be attributed to an enhanced, dataset-specific feature set and a relatively small, well-structured dataset, which improved class separability and kernel effectiveness. Our QML experiments, conducted on PCA-reduced features from a larger and more overlapping dataset, faced greater classification complexity, limiting achievable performance. This suggests that quantum models are most effective in structured, lower-dimensional, and well-separated feature spaces, and may benefit from hybrid classical–quantum pipelines.
Finally, this work does not evaluate real quantum hardware. All QML experiments were performed using Pennylane-based simulations, providing a realistic benchmark under near-term hardware constraints rather than demonstrating performance on physical quantum devices.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of ASD traits across different demographic factors within the dataset.

Table 4: Comparison with Previous ASD Prediction Models (Classical + Quantum)

Study	**Algorithm**	**Acc.**	**Prec.**	**Rec.**	**F1**
A Comparison of Machine Learning Algorithms for the Surveillance of Autism Spectrum Disorder [5]	Naive Bayes	73	–	–	78
	CVM	87	–	–	83.6
	Random Forest	87.1	–	–	86.8
An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder	Random Forest	94	90.1	88.3	–
	Naive Bayes	85	84	81	–
	SVM	80	81	80	–
Machine Learning-Based Models for Early Stage Detection of Autism Spectrum Disorders	Naive Bayes	96	–	–	97
	SVM	85	–	–	–
	KNN	90	–	–	–
	Random Forest	**99**	–	–	–
Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning	Logistic Regression	97.15	–	–	98
	Naive Bayes	94	–	–	96
	KNN	90.52	–	–	93
	Random Forest	81.52	–	–	88
Quantum-Based Machine Learning Approaches — Saranya & Menaka (2025) [4]	QSVM (with enhanced feature set)	98.9	–	–	–

Distribution Based on Ethnicity ASD traits were found to be most prevalent in White Europeans within this dataset (Fig 12). This observation could be influenced by various factors, including genetic, social, environmental influences, and potentially higher awareness and access to diagnostic testing in these populations.

Distribution Based on Sex: Analysis indicated that males exhibited ASD traits more frequently than females in this dataset (Fig 13). Hormonal differences, such as the influence of testosterone on brain development, are often considered as potential contributing factors to this observed gender disparity in ASD prevalence.

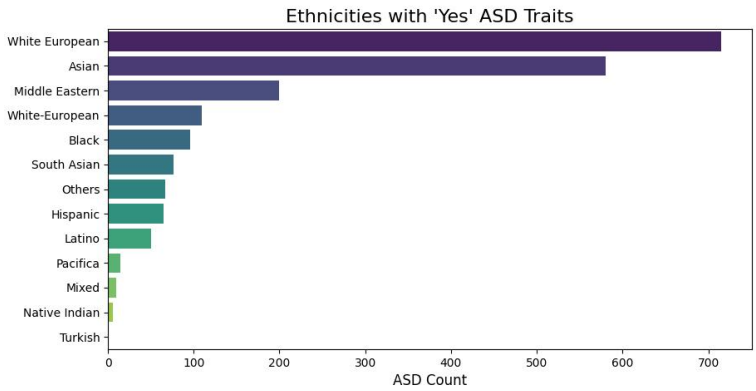

Fig. 12: The ethnics with the ASD traits as "YES"

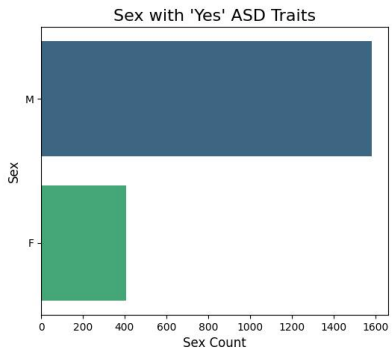

Fig. 13: Gender distribution with ASD traits as "YES"

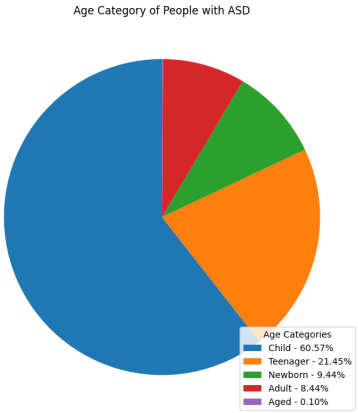

Fig. 14: Age categories distribution of ASD traits:

Distribution Based on Age Category It was observed that ASD traits were most commonly reported in children, specifically affecting 60.57% of individuals in the child age category (2-12 years) within the dataset (Fig 14, Table 5). This highlights a period of significant susceptibility to exhibiting these traits.

5 Conclusion

This study systematically compared classical supervised machine learning models with Quantum Machine Learning (QML) approaches for ASD trait prediction on non-linear datasets. Among classical methods, the Histogram Gradient Boosting Classifier (HGBC) achieved the strongest

Table 5: Age Category Traits in Percentage

Age Category	**Traits (%)**
Newborn	9.44%
Child	60.57%
Teenager	21.45%
Adult	8.44%
Aged	0.10%

overall performance, reaching 97.86% accuracy and consistently outperforming other baselines.
Within the quantum models, both Quantum Support Vector Machine (QSVM) and Quantum Feature Map for Naive Bayes (Q→NB) demonstrated competitive results, with Q→NB achieving 93.24% accuracy and QSVM achieving 82.00%, while the Variational Quantum Classifier (VQC) showed comparatively lower performance (74.50%). These results indicate that, with optimized quantum encodings and suitable hardware, QML approaches hold potential for modeling complex non-linear relationships even at smaller scales.
Exploratory Data Analysis further revealed higher ASD trait prevalence in children (60.57%) and males, along with demographic patterns related to ethnicity and family history, highlighting the multifactorial nature of ASD.
Overall, classical ensemble models such as HGBC and Random Forest remain the most reliable under current computational constraints. However, quantum-based approaches present a promising direction, particularly without dimensionality reduction. Future work will focus on larger datasets, richer clinical feature sets, hybrid classical–quantum architectures, and further hyperparameter optimization of kernel-based quantum models to better compete with state-of-the-art classical methods.

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